HIRONAKA'S EXAMPLE OF A PROPER MORPHISM OF SMOOTH COMPLEX THREEFOLDS THAT IS ZARISKI-LOCALLY PROJECTIVE BUT NOT PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Hironaka example; the companion lemma lemma-non-effective-descent-projective reuses this construction and its equivalence conclusion.

Editorial note. The label lemma-non-descending-property-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly failed at compile time (the author's citation of Hironaka via the bare key H had no bibliography backend in r001; the stale r001 job directory is retained as evidence).

Editorial note. Bibliography support: the counted proof cites the letter-key H (the Stacks Project's global bibliography entry for Hartshorne, Algebraic Geometry, Graduate Texts in Mathematics 52, Springer, 1977, whose Appendix B Example B.3.4.1/B.3.4.2 are the Hironaka examples being imitated). A matching hironaka.bib asset is attached so the author's own citation resolves; no statement or proof text was altered.

Allowed standard prerequisites (editorial).

- Divisors, Definition divisors-definition-strict-transform: the strict transform of a closed subscheme under a blowup
- Divisors, Lemma divisors-lemma-blowing-up-projective: a blowup morphism is projective
- Chow Homology, Lemma chow-lemma-proper-pushforward-rational-equivalence and Lemma chow-lemma-factors: the degree of a line bundle on a proper scheme over a field is well defined on one-cycles modulo rational equivalence
- The conics C given by x y equals z squared and w equals zero, and D given by x y equals w squared and z equals zero, in projective three-space over the complex numbers are smooth, meet transversally in the two reduced points P equals the point with coordinates one zero zero zero and Q equals the point with coordinates zero one zero zero, and each is isomorphic to the projective line

Source reference. Original lemma 02TI: sources/stacks-chow/tags.txt, source lines 3697–3697. Prerequisite pointer invoked at 5434; it names a standard fact of a

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sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02S2: sources/stacks-chow/tags.txt, source lines 3645–3645. Prerequisite pointer invoked at 5433; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 080D: sources/stacks-chow/tags.txt, source lines 10178–10178. Prerequisite pointer invoked at 5398; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02NS: sources/stacks-chow/tags.txt, source lines 3491–3491. Prerequisite pointer invoked at 5405; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The properties*

$\mathcal{P}(f) =$ *"f is projective", and*
$\mathcal{P}(f) =$ *"f is quasi-projective"*

are not Zariski local on the base. A fortiori, they are not fpqc local on the base.

Proof. Following Hironaka [1, Example B.3.4.1], we define a proper morphism of smooth complex 3-folds $f : V_Y \to Y$ which is Zariski-locally projective, but not projective. Since f is proper and not projective, it is also not quasi-projective.

Let Y be projective 3-space over the complex numbers $\mathbf{C}$. Let C and D be smooth conics in Y such that the closed subscheme $C \cap D$ is reduced and consists of two complex points P and Q. (For example, let $C = \{[x, y, z, w] : xy = z^2, w = 0\}$, $D = \{[x, y, z, w] : xy = w^2, z = 0\}$, $P = [1, 0, 0, 0]$, and $Q = [0, 1, 0, 0]$.) On $Y - Q$, first blow up the curve C, and then blow up the strict transform of the curve D (Divisors, Definition 080D). On $Y - P$, first blow up the curve D, and then blow up the strict transform of the curve C. Over $Y - P - Q$, the two varieties we have constructed are canonically isomorphic, and so we can glue them over $Y - P - Q$. The result is a smooth proper 3-fold V_Y over $\mathbf{C}$. The morphism $f : V_Y \to Y$ is proper and Zariski-locally projective (since it is a blowup over $Y - P$ and over $Y - Q$), by Divisors, Lemma 02NS. We will show that V_Y is not projective over $\mathbf{C}$. That will imply that f is not projective.

To do this, let L be the inverse image in V_Y of a complex point of $C - P - Q$, and M the inverse image of a complex point of $D - P - Q$. Then L and M are isomorphic to the projective line $\mathbf{P}^1_{\mathbf{C}}$. Next, let E be the inverse image in V_Y of $C \cup D \subset Y$ in V_Y; thus $E \to C \cup D$ is a proper morphism, with fibers isomorphic to $\mathbf{P}^1$ over $(C \cup D) - \{P, Q\}$. The inverse image of P in E is a union of two lines L_0 and M_0, and we have rational equivalences of cycles $L \sim L_0 + M_0$ and $M \sim M_0$ on E (using that C and D are isomorphic to $\mathbf{P}^1$). Note the asymmetry resulting from the order in which we blew up the two curves. Near Q, the opposite happens. So the inverse image of Q is the union of two lines L'_0 and M'_0, and we have rational equivalences $L \sim L'_0$ and $M \sim L'_0 + M'_0$ on E. Combining these equivalences, we find that $L_0 + M'_0 \sim 0$ on E and hence on V_Y. If V_Y were projective over $\mathbf{C}$, it would have an ample line bundle H, which would have degree > 0 on all curves in V_Y. In particular H would have positive degree on $L_0 + M'_0$, contradicting that the degree of a line bundle is well-defined on 1-cycles modulo rational equivalence on a proper scheme over a field (Chow Homology, Lemma 02S2 and Lemma 02TI). So V_Y is not projective over $\mathbf{C}$. □

References

[1] Robin Hartshorne. *Algebraic Geometry.* Number 52 in Graduate Texts in Mathematics. Springer-Verlag, New York-Heidelberg, 1977.